Shaan Patel

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Objective

Ambitious third-year Computer Engineering student specializing in RTL design, low-latency FPGA design, and micro-architectures. Seeking an FPGA engineering internship to apply expertise in SystemVerilog RTL implementation, networking hardware integration, and EDA synthesis and tool flows to create high performance silicon systems.

Education

Georgia Institute of Technology — Atlanta, GA

Bachelor of Science in Computer Engineering — *August 2023 – May 2027*

Master of Science in Electrical and Computer Engineering — *August 2027 – May 2028*

- Minor: Applications of Artificial Intelligence and Machine Learning | GPA: 3.57
- Relevant Coursework: Data Structures and Algorithms, Embedded Systems Design, Advanced Microarchitecture Design
- Leadership: Captain of GT Ramblin' Raas, GTXR Executive Board, SiliconJackets Digital Design Team Lead

Technical Skills

Hardware: FPGA, ESP32, I2C, SPI, IC Breadboard Circuits, Oscilloscope, Arduino, Raspberry Pi, ARM MBED, Logic Analyzer
Verification: UVM, SVA, Functional Testing , Coverage Analysis, RTL Simulation & Debug, Logic Synthesis, Place & Route (P&R)
Programming: C++, SystemVerilog, C, VHDL, RISC-V, Verilog, OpenGL, Python, Java, JavaScript, Tcl, Perl, Shell Scripting
Platforms: Linux (Ubuntu, Slackware, Debian), Red Hat (OpenStack, CloudForms), OpenVMS, Solaris, Windows, MacOS
Frameworks: React, SFML, CUDA, PyTorch, AWS (S3, DynamoDB, EC2), Express, MongoDB, Pandas, Node.js, Kubernetes
Software: KiCAD, Altium, Cadence Virtuoso, Intel Altera Quartus II, LTspice, Verilator, gRPC, Blender, Autodesk Fusion 360/AutoCAD, SolidWorks, Onshape, Visual Studio Code, GitHub, Wireshark, Adobe Creative Suite, Inkscape, Slack

Experience

AeroCore-V – Hardware-Accelerated SoC for Autonomous UAVs — Atlanta, GA

SiliconJackets Project Lead — *August 2025 – Present*

- Designed RTL for an open-source RISC-V SoC in SystemVerilog with custom single-cycle ALU extensions for single-cycle PID control math to achieve low-latency, nanosecond response times for UAV flight loops
- Integrated a direct-mapped, 4KB L1 cache handling high-frequency I2C/SPI GPS and IMU data streams to reduce stalls by 52% (measured via a custom Verilator cycle-accurate harness), while developing a 4-way L2 controller with LRU replacement
- Developing a virtual memory system with Page Table Walker and TLB to isolate flight routines and protect from page faults, utilizing Github Actions CI/CD pipelines to automate regression testing and cycle-accurate simulations
- Prototyped a priority-based thread scheduler to manage UAV flight loops and telemetry tasks, alongside an OpenGL Digital Twin physics simulator in C++ to validate future flight latency reductions
- Validated and targeted high-speed hardware designs on Intel DE10 FPGAs with Intel Quartus Prime and authored SystemVerilog Assertions (SVA) and functional testbenches in Verilator to perform coverage analysis and debug RTL logic failures

Georgia Tech HIVE Makerspace — Atlanta, GA

Student Leader and Volunteer — *August 2024 – Present*

- Instruct ∼40 students per week on advanced design and fabrication workflows, including 3D CAD/slicing, circuit analysis with oscilloscopes, laser cutter operation, PCB design and manufacturing (taking a blank FR4 sheet to an operational PCB)
- Deployed an open-source multi-device print manager to automate jobs on a FIFO basis, increasing the operational uptime of the free-to-use 12-unit Bambu Lab 3D print farm for the Georgia Tech student body
- Facilitate workshops and created training resources that contributed to a 20% increase in student participation
- Architecting a full-stack inventory management system using an AWS RDS backend to track materials in the cloud with S3 and DynamoDB to enable a student electronics rental program, aiming to reduce annual student costs

High-Performance Heterogeneous Computing Infrastructure — Ashburn, VA

BlendFarm Project Leader — *June 2022 – July 2025*

- Assembled two 2.5kW parallel computing systems from the component level, integrating twelve 12GB GDDR6 GPUs and modifying voltage curves to bypass LHR limiters, achieving a 73% hash rate improvement
- Overcame data mesh faults using Tcl and Bash automations to schedule distributed compute workloads across cluster nodes
- Scripted a distributed render farm using C++ and SFML to coordinate large ray tracing frame rendering across a network
- Optimized low-level networking routines via TCP/UDP socket interfaces and a custom jitter buffer implementation to ensure reliable data synchronization between the master node and rendering clients for faster computer graphics generation

Academy of Science International Research — Ashburn, VA

Research Collaborator — *November 2022 – July 2024*

- Synthesized a novel saline-soluble polymer alternative to general petrochemical plastics as a means of preventing marine pollution
- Modeled and fabricated 3D-generated molds using filament and resin printers for synthesizing hydrogels through incubation
- Led collaboration efforts with South Korean research teams for two years to formulate samples and complete testing
- Collected over 150 trials of data and was awarded the Yale Outstanding Project award at RSEF in Spring of 2023